Stepan



Making hair Naturally Beautiful

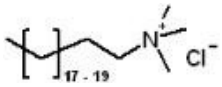
Bio-based Ester Quats & Emollients

Ester quats in hair conditioners

Substitutes for BTAC/CTAC

BTAC, CTAC, EQ

BTAC (behentrimonium chloride)



CTAC (cetrimonium chloride)

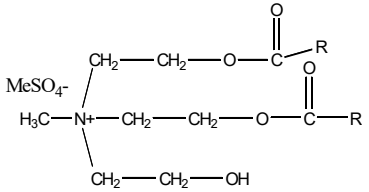


- BTAC: solid form, blended with fatty alchohol or IPA
- CTAC: liquid form, in a solvent (water or IPA)
- Conditioning & anti-static agents, bring smoothness & shine
- **Restricted since 08/2014 in rinse-off hair products & leave-on hair & face products**
- CLP classification of PURE SUBSTANCES :



Stepan Ester Quats

Dipalmitoylethyl Hydroxyethylammonium Methosulfate



- Paste or liquid form, several solvents possible
- Cost effective conditioning agents
- No use restriction
- Eco and user friendly product
- Pure substance non CLP classified

Our offer

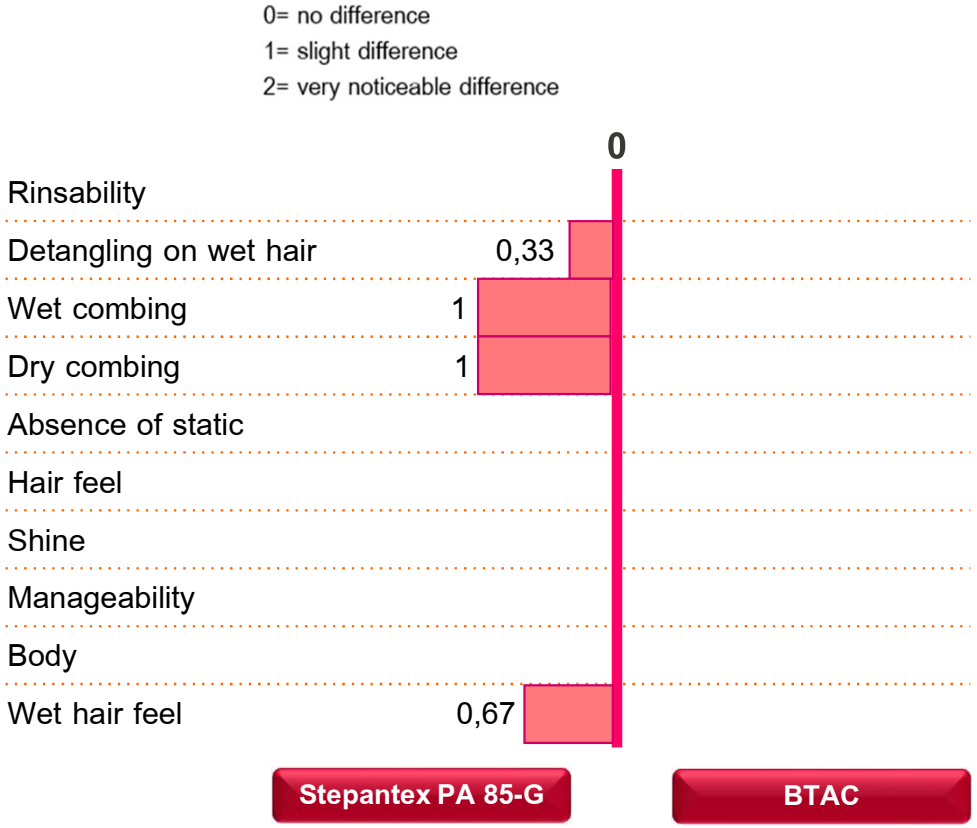
	STEPANQUAT GA 90	STEPANTEX PA 85-G
INCI	Dipalmitoylethyl Hydroxyethylammonium Methosulfate (China approved)	
Physical form	Paste	
Solvent	IPA	DPG
Key benefits	Good ecotox profile No restriction No labelling on active substance	
Downsides	Flammable Paste form	Paste form

Efficient alternatives

Equivalent to BTAC

Half-head tests (3 panelists)

	% actives
Stepanquat PA85G (85%) or BTAC	2
Cetyl alcohol	3
Stearyl alcohol	1
Steareth-20	0,5
DI water	93,15



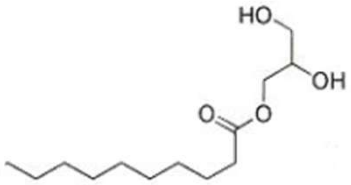
Conditioning boosters

Bio-based emollients

Conditioner boosters

Stepan-Mild® GCC

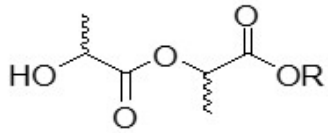
Glyceryl Caprylate/Caprate





Stepan-Mild® L3

Lauryl Lactyl Lactate



R = Predominantly C_{12}

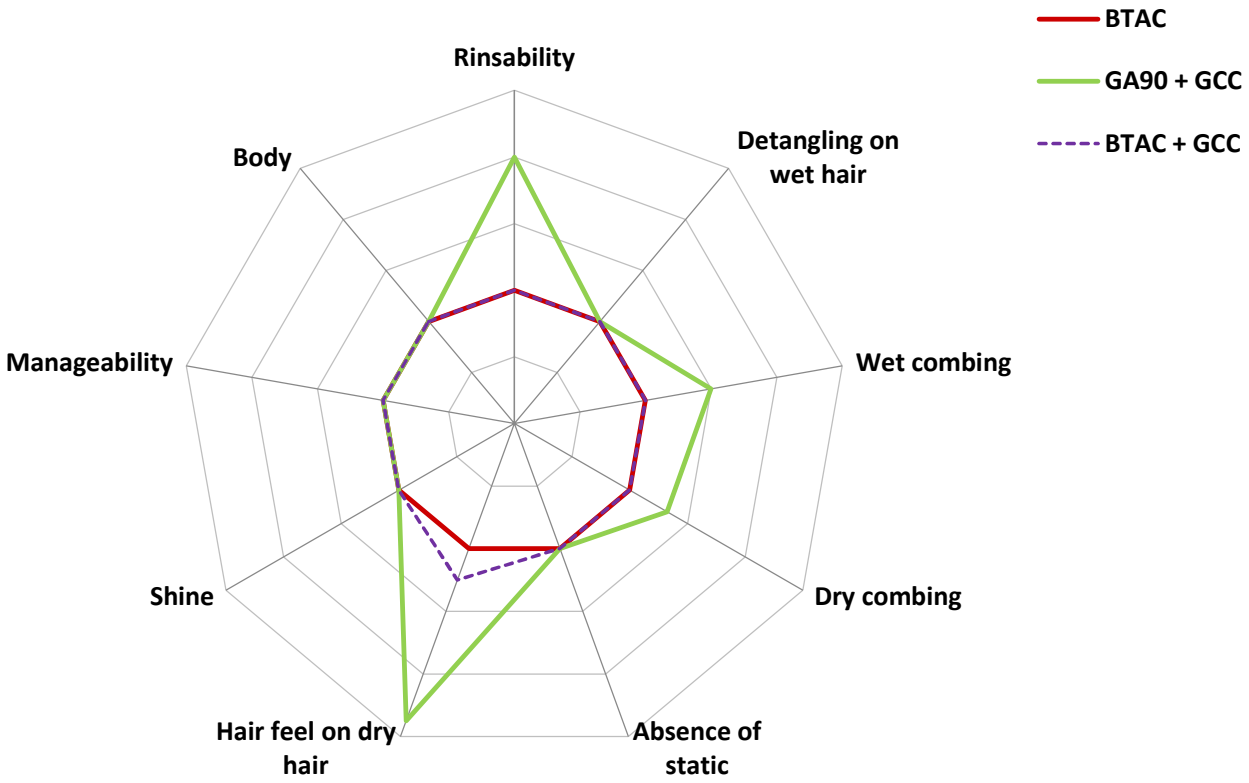


100% actives, 100% bio-based

Fatty chain = VdW interactions with hair cuticle's fatty acids

Stepan-Mild® GCC

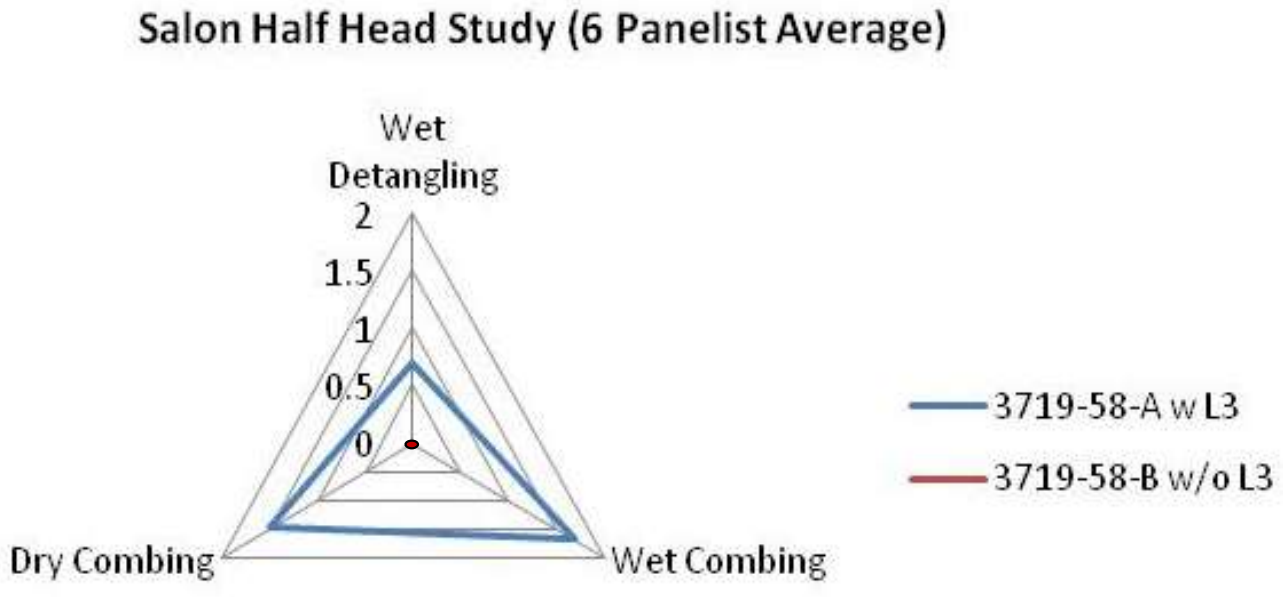
Preliminary sensorial hair test



Preliminary sensorial test
Done by Technical HairDresser
5 panelists

STEPANQUAT®GA 90
in combination with
STEPAN-MILD® GCC
give a close match to BTAC.

Stepan-Mild® L3



Preliminary sensorial test

Done by Technical HairDresser

6 panelists

Hair Conditioner with *STEPAN-MILD® L3*

shows improvement

in wet and dry combing,

slight improvement in wet detangling.

0= no difference

1= slight difference

2= very noticeable difference

Hair conditioner formulation

Ingredients	% active	%w/w
DI Water		Qs. 100
Stepanquat® GA90 Dipalmitoylethyl Hydroxyethylammonium Methosulfate	2	2.22
Stepan-Mild® L3 Lauryl Lactyl Lactate	up to 1	1
Stepan-Mild® GCC *Glyceryl Caprylate/Caprate*	up to 1	1
Cetyl alcohol	3	3
Stearyl alcohol	1	1
Steareth-20	0.5	0.5
pH adjuster, preservative, perfume		Qs. to pH = 4



Cleansing conditioner formulation

Ingredients	% active	%w/w
DI Water		Qs. 100
Ammonyx® LO Lauramine Oxide	6	20
Stepan-Mild® GCC Glyceryl Caprylate Caprate	6	6
Stepan-Mild® L3 Lauryl Lactyl Lactate	1	1
Stepanquat® GA90 Dipalmitoylethyl Hydroxyethylammonium Methosulfate	2	2,22
Decyl Glucoside	2	4
Cocamide MIPA	0,8	0,8
Sunflower oil	4	4
Jojoba Oil	1	1
Perfume, preservative		qs
Lactic acid		Qs. to pH < 4,5



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